

2(a)	Gain in kinetic energy = loss in gravitational potential energy $\frac{1}{2}mv^2 = mgh$ $v^2 = (10.0)(9.81)2$ $v = 14.0 \text{ m s}^{-1}$ $\cos \theta = \frac{30-10}{30} = \frac{2}{3}$ $\theta = 48.2^\circ$	M1 A0 M1 A0
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2b(i)	The path is a parabola. A constant vertical force, weight, acts on the bolt while its horizontal velocity remains constant as there is no horizontal force. [Reference to forces acting on the ball must be made. Avoid using the term 'gravity']	B1 B1
2b(ii)	$s_y = u_y t + \frac{1}{2} a_y t^2$ $20.0 = 14.0 (\sin 48.2^\circ) t + \frac{1}{2} (9.81) t^2$ $t = 1.22 \text{ s}$	C1 A1
(c)	The graph shows vertical acceleration a_y on the y-axis and time t on the x-axis. The curve starts at the origin and increases with a decreasing gradient, becoming a horizontal red line at time B. Vertical dashed lines mark 'time at A', 'time at B', and 'time at C'.	B1
(d)	The graph shows vertical velocity v_y on the y-axis and time t on the x-axis. The curve starts at the origin and increases with an increasing gradient, becoming a straight red line at time B. Vertical dashed lines mark 'time at A', 'time at B', and 'time at C'. A to B: curve up showing increasing velocity with increasing gradient from zero to maximum (B1) B to C: straight line showing increasing velocity with constant gradient (B1)	